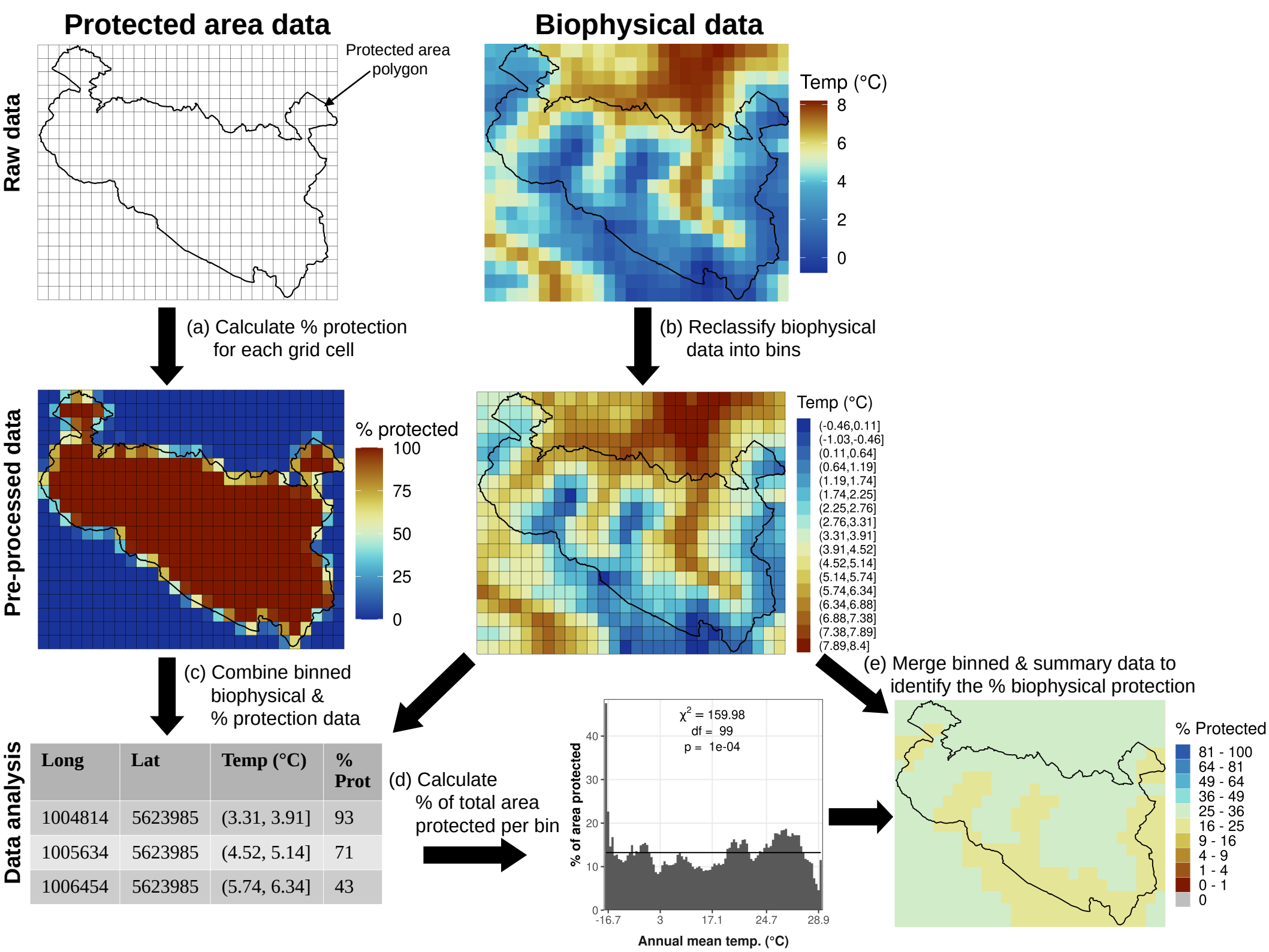
Protected area data
Biophysical data
Raw data
Protected area polygon
Temp (°C)
8
6
4
2
0
(a) Calculate % protection for each grid cell
(b) Reclassify biophysical data into bins
Pre-processed data
% protected
100
75
50
25
0
Temp (°C)
(-0.46,0.11]
(-1.03,-0.46]
(0.11,0.64]
(0.64,1.19]
(1.19,1.74]
(1.74,2.25]
(2.25,2.76]
(2.76,3.31]
(3.31,3.91]
(3.91,4.52]
(4.52,5.14]
(5.14,5.74]
(5.74,6.34]
(6.34,6.88]
(6.88,7.38]
(7.38,7.89]
(7.89,8.4]
(c) Combine binned biophysical & % protection data
(e) Merge binned & summary data to identify the % biophysical protection
Data analysis
Long | Lat | Temp (°C) | % Prot
1004814 | 5623985 | (3.31, 3.91] | 93
1005634 | 5623985 | (4.52, 5.14] | 71
1006454 | 5623985 | (5.74, 6.34] | 43
(d) Calculate % of total area protected per bin
χ² = 159.98
df = 99
p = 1e-04
% of area protected
40
30
20
10
0
-16.7
3
17.1
24.7
28.9
Annual mean temp. (°C)
% Protected
81 - 100
64 - 81
49 - 64
36 - 49
25 - 36
16 - 25
9 - 16
4 - 9
1 - 4
0 - 1
0